

Gnome 100 horsepower

To run satisfactorily on the poor-quality fuel available at the beginning of the 20th century, early aero engines had to operate at a low compression ratio, which caused them to run at high temperatures. This virtually ruled out the viability of air-cooled units until the appearance, in 1908, of the Seguin brothers' ingenious Gnome "rotary." Rotary engines remained unrivaled in power-to-weight ratio until the end of World War I.

ROTARY DESIGN

While it may look like a classic radial, the 100 hp Gnome engine is radically different in operation. Instead of the engine being mounted on the airframe and the propeller being driven by the crankshaft, the rotary's crankshaft is bolted rigidly to the aircraft and the entire engine spins as one unit with the propeller. The beauty of this arrangement was that the cylinders were subjected to high-speed airflow at the low flying speeds typical of the era, and the "flywheel" effect helped keep the engine running. The improved Monosoupape (single-valve) took Gnome reliability to a higher level, ultimately producing 160 hp.

ENGINE SPECIFICATIONS	
Dates produced	1914-18
Configuration	Air-cooled 9-cylinder rotary
Fuel	Gasoline
Power output	110 hp @ 1,100 rpm
Weight	297 lb (135 kg)
Displacement	993 cu in (16.3 litres)
Bore and stroke	4.9 in (124 mm) x 5.9 in (150 mm)
Compression ratio	5.5:1



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Front view
Like all four-stroke rotaries and radials, the Gnome had an odd number of cylinders to give even firing intervals. The finely machined cooling fins were made deeper where greater cooling was required.

Spark plug (one per cylinder)
Unlike later aero engines, this early Gnome engine relied on single ignition.

Cooling fins
Closely spaced, these were beautifully designed for optimal cooling.

Propeller mount
The propeller rotated with the engine.

Ignition wire anchor ring
The current from a stationary magneto mounted on the bulkhead was carried to each plug in turn by uninsulated wires.

Side view
From this angle the exposed rocker gear and exhaust valve at the top of each cylinder is obvious. Castor oil lubricant was sprayed out with the exhaust, to the detriment of airframe and pilot alike.

